GROUP 22 | MACHINE LEARNING MINI-PROJECT

STUDENT STRESS LEVEL PREDICTION

Predicting students' perceived stress using social media usage and sleep patterns

TEAM MEMBERS & RESPONSIBILITIES

ANANYA: 2026MCB1510
Documentation Lead and Coding Lead

AVDIT BASERA: 2026AIB1024
Video Lead

HARKEERAT AGGARWAL: 2026DAB1293
Execution Lead

KHARATMOL SHALINI: 2026EEB1365
PPT Lead

PROJECT REPOSITORY: https://github.com/Ananya9-tech/AI101-Mini-Project.git

PROBLEM & DATASET

Use daily social media usage and sleep patterns to estimate a student's perceived stress score.

THE PROBLEM

Can daily social media usage, sleep duration, and sleep quality be used to predict a student's perceived stress score?

The goal is to estimate a numerical stress score from selected daily-use and sleep-related features.

TARGET (y): Perceived_Stress_Score

DATASET & INPUT FEATURES

4,500 rows | **16 columns** | **4 input features**

INPUT FEATURES (X)

- Daily_Usage_Hours
- Sleep_Duration_Hours
- Sleep_Quality_Score
- Weekend_Extra_Hours

METHODOLOGY

A regression work flow uses four selected features to predict one numerical stress score.



TOOLS & LIBRARIES

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Train–test split: 75% training data & 25% testing data

1,125 test rows assumes all 4,454 dataset rows are retained after preprocessing.

INITIAL MODEL RESULTS

Linear Regression • Initial run: 4454 training samples and 192 test samples

R^2 SCORE

Training R2: 0.593
Testing R2: 0.615

MAE

Training MAE: 3.555
Testing MAE: 3.668

RMSE

Training RMSE: 4.433
Testing RMSE: 4.604

WHAT THE RESULTS WILL SHOW

- How closely the predicted stress scores match the actual scores.
- The average prediction error and the size of larger prediction errors.

LEARNINGS & TEAM CONTRIBUTIONS

What we completed, what remains to verify, and how the team contributed..

KEY LEARNINGS

- Identified the social media and sleep dataset.
- Selected four input features and one target variable.
- Linear Regression is the planned prediction model.
- Final model metrics still need to be generated and checked.

